SPIRE

A Contested-Logistics Operating System

SPIRE collapses the data layer underneath military logistics into one role-shaped view. It runs offline on a 2U box in a CONEX, with on-board AI, tamper-evident audit, and one-click coalition release. Built by serving Marines for joint and partnered force employment in degraded, disconnected, intermittent, and limited (DDIL) communications.

TRL 4. 1st place at Modern Day Marine 2026.

Four operator surfaces

Decision Bridge — the home view

The first screen a Marine sees when they sign in. Five live signals on one non-scrolling pane, none of them deeper than two clicks: the mission clock and the current force-protection condition; the top three alerts by severity drawn from every active feed; the next forecasted shortages across Class I, V, VIII, and IX with hours-to-impact projected from the readiness model; the mission-capable rate by unit with a seven-day sparkline showing trend direction; and the health of the audit chain itself — entries per minute, last hash, integrity status. Auto-refresh runs at five to sixty seconds depending on the tile. An air-gap-degraded indicator fades the live tiles when SATCOM drops so the operator immediately sees their picture is no longer fresh.

PULSE — readiness and risk

Mission-capable rate by unit by equipment type rendered as a heatmap so a maintenance chief sees at a glance which combination is bleeding. Monte Carlo demand forecasting projects 200 stochastic paths across the next fourteen days, surfacing the p10 / p50 / p90 envelope on Class IX requisitions. When a shortage forecasts past the readiness threshold, PULSE auto-drafts ranked options for the planner — cannibalize from a low-mission unit, expedite from depot, cross-level from a peer — sorted by impact-per-day. Predicted equipment failures use a mean- time-between-failures model per equipment type to project component failures days ahead and draft the requisition before the asset breaks. Every action is signed and chained to the audit log; the decision the S-4 made at H+0 is verifiable against the public key at H+30 days.

BASTION — common operating picture

Real geographic map tiles with standard military symbols, not hand-drawn icons. Multi-sensor threat fusion correlates access-control gate events, drone detections, base-utility (SCADA) anomalies, and weather advisories into single fused-threat records with confidence scores and visual correlation chains an operator can scrub through. QRF cordons render dynamically when an incident triggers; threat rings overlay at configurable ranges; unit symbols update from the live roster. A natural-language copilot accepts queries like "fly to MCAS Iwakuni" or "show 3d MLR units in scope" and translates them into map operations. The operator never has to learn a coordinate system or a map-control vocabulary.

SENTRY — classification and release

A six-stage pipeline with operator override at every step: Upload, Processing, Review Queue, Mark Draft, Export, and Coalition. Tier-1 rule-based classification runs synchronously and deterministically against a configurable rule set — every input gets the same answer. Tier-2 explanations are opt-in: an operator clicks "explain" and an on-board AI model writes a paragraph anchored on the Tier-1 evidence spans, naming the rules that fired and proposing redacted phrasing the operator can paste back. Bulk uploads route through a sanitization gate that rejects clear PII before it enters the canonical store. Sanitized exports build a ZIP with the manifest hash, the bundle hash, and a snapshot of the audit chain entries that wrote the records.

The platform underneath

Universal Ingest — any file, any source

Reads CSV, tab-separated values, Excel workbooks, JSON, JSON-Lines, XML, fixed-width legacy exports, and EDI X12 supply transactions out of the box. Each format gets the right parser without operator configuration — the system infers what it's reading from the first kilobyte. Six source channels cover every common DoD ingest pattern: drag a file in directly, watch a local directory for new arrivals, pull from a remote SFTP share on a schedule, watch an inbox for attachments matching configured filters, poll a REST or SOAP web service with ETag-aware caching, replay change-data-capture events from an Oracle / PostgreSQL / SQL Server source database, or subscribe to a Kafka streaming topic with offset persistence. Every channel carries retry-with-backoff, circuit breakers, byte-bounded fetch caps, and a dead-letter queue so one bad payload doesn't poison the rest of the pipeline.

Pre-mapped Marine logistics on day one

The four systems a Marine logistics shop touches every day arrive ready to ingest — GCSS-MC asset roster (ECP), GCSS-MC utilization extract (UTIL), GCSS-MC service-request header (SR-Header), and DRRS-MC unit readiness ratings (C-Rating). Each adapter declares its column types, sensitivity flags, primary keys, fallback keys, and cross-field constraints in a single declarative spec. No buried logic, no contractor-only knowledge. Extending the catalog to GCSS-Army, DLA EBS, TC-AIMS, or any other tabular system is a one-day engineering task per adapter.

On-board AI for unfamiliar files

When an operator drops a non-canonical file — a unit-specific spreadsheet, a partner-provided extract, a one-off report from a contractor — a language model running entirely on the same 2U box reads the headers and proposes how each column should map to the canonical schema. Sample data is scrubbed of PII before the model sees it; no row leaves the box. The operator confirms the proposal once; the mapping saves as a per-unit reusable profile. The unit doesn't redo the work next time the same shape arrives.

Tamper-evident audit chain

Every change to canonical state — every record applied, every classification mark, every coalition release, every operator approval — gets a hash entry chained to the previous entry. High-value entries are also signed with an Ed25519 key so they verify offline against a published public key, even if the underlying database file is rewritten by an attacker with disk access. Compliant with NIST 800-53 AU-9(5). The audit log forwards in Common Event Format over UDP or TCP to whatever security operations center the host installation runs — Splunk, ArcSight, QRadar, or a downstream syslog collector. No proprietary format, no vendor lock.

Coalition release — five partners ship pre-built

Five-Eyes, Japan, Australia, the Philippines, and a NATO-base profile. Each profile carries its own field-level redaction rules, classification ceiling, embargo windows, and authorized unit set. The redaction engine applies live to every record at release time. The operator sees the diff — what each partner will see versus what's in the canonical store — before the bundle leaves the box. The receiving partner verifies the bundle hash on receipt against the signature stamped to the SPIRE audit chain. Generating a partner-scoped release is one click after the operator picks the profile.

Identity hardening

CAC/PIV certificate-based authentication is built to production shape in mock-PIN mode by default; a single environment-variable flip moves the system to require real DoD certificate forwarding from the TLS-terminating proxy. The EDIPI extracts from the certificate's subject Common Name; the chain validates against trust anchors loaded from a configurable directory; revocation modes for OCSP and CRL are wired in as scaffolds awaiting deployment-time configuration. Hierarchical role-based access maps to the joint logistics command chain: a G-4 at CLR sees every CLB beneath, a SSgt at CLB doesn't see peer CLB data, write access is locked to the operator's home unit. Records retention is DoD 5015.02-compliant with hard-delete capability for spillage cleanup, and every retention deletion gets its own audit chain entry — kind, filename, hash, age, reason — so an investigator can reconstruct what was deleted without recovering the file. Secrets (channel credentials, signing keys, partner API tokens) reference HashiCorp Vault via URL; the Vault adapter fetches at process start and never persists resolved secrets to disk.

Local-first, no cloud, ever

The whole stack — backend, frontend, database, on-board AI model, audit chain — runs on a single 2U server in a CONEX with no outbound network connectivity required. When a forward node and a rear node both write during a SATCOM blackout, vector-clock reconciliation merges their state on link restore: writes that don't conflict apply automatically; writes that do conflict surface to a Security Manager for resolution, with the audit chain preserving both the winner and the loser of every conflict. Forward nodes are first-class, not read-only replicas.

What this lets a force do

A forward operator sees the unit's posture across Class I, V, VIII, and IX on one screen instead of three browser tabs and a phone call. A planner gets ranked options before the asset breaks, not after the work order opens. A G-4, J-4, or S-4 sees the same canonical shape regardless of which service's data flows through the pipe. A coalition partner gets exactly what their release authority allows, signed and verifiable. An inspector walks the audit chain offline against a public key. The whole thing runs without satellite, without cloud, on a single 2U box. From a clean laptop with Docker installed, `docker-compose up` brings the whole stack live in under five minutes — no registration, no license server, no cloud dependency.

Built to IL5 standards (tamper-evident audit, role-based access, smartcard authentication, no cloud egress). Not formally accredited.

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Repo: github.com/jeranaias/spire (public) · `docker-compose up` — clean laptop to live demo in under 5 minutes